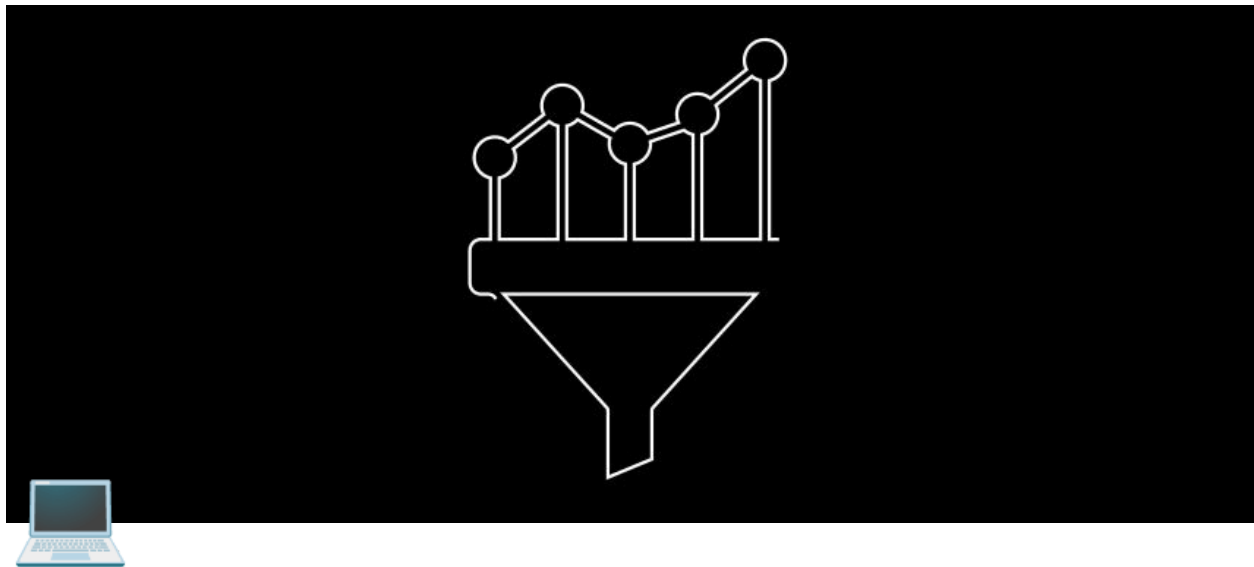




Funnel Analysis (1)

☰ Tags	EDA Excel Excel Solver Python
≡ Description	Customer Journey Funnel Optimization and Resource Prioritization

Executive Summary

Approximately **20.2% of users complete a purchase**, with the largest friction occurring at the **Product Page → Add to Cart** stage. Funnel bottlenecks are consistent across continents, indicating **structural issues rather than regional differences**.

Conversion efficiency varies by time of day, with **Afternoon users being the most likely to convert. In contrast**, evening traffic generates the highest volume but underperforms in conversion. Late-night traffic is smaller but converts relatively well.

Overall findings suggest **product page clarity, mobile usability, and perceived value** as the primary levers for improving conversion.

Problem

Despite strong traffic volume, many customers fail to complete purchases. The objective was to identify where customers disengage, identify potential underlying

causes, and recommend improvements that enhance customer experience and, in turn, conversion.

Analytical Approach

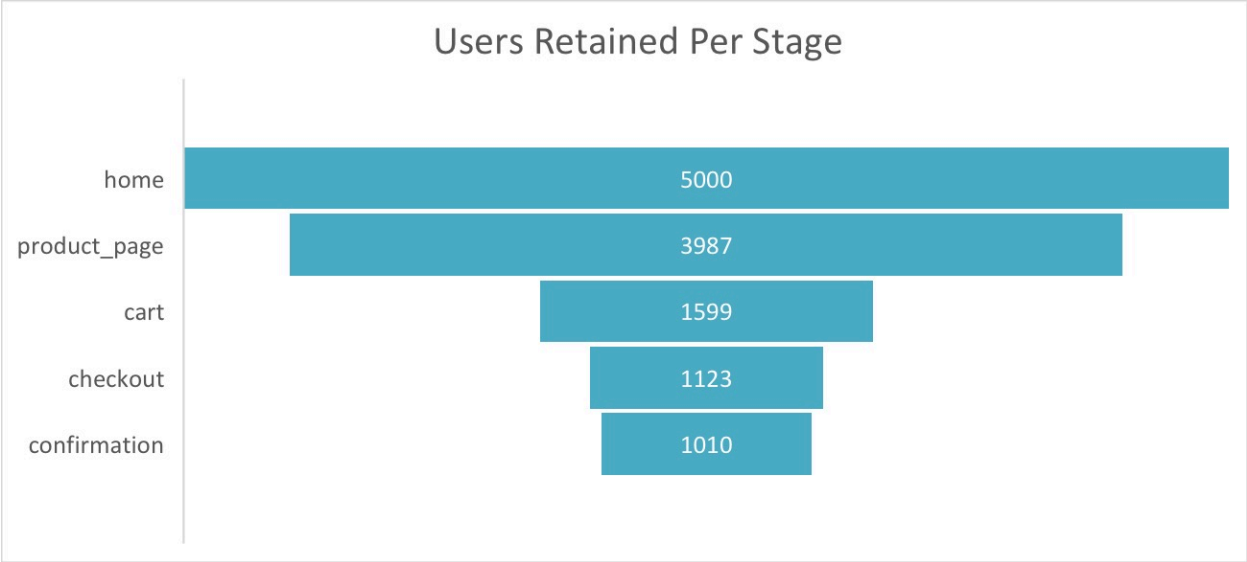
Event-level session data was processed to reconstruct a multi-stage conversion funnel from landing page to purchase confirmation. Users were segmented by **device type**, **geography**, **referral source**, and **time of day** using descriptive and comparative analysis

Technical Implementation

- **Python** for:
 - Converting CSV → XLSX
 - Data cleaning and validation
 - Exploratory analysis and aggregation using SQL queries in Python
- **Excel** for:
 - Creating pivot tables from Python-generated tables
 - Building charts and visualizations from pivot tables

Automation of the workflow reduced preparation time by **60–70%**, enabling faster iteration and reproducibility.

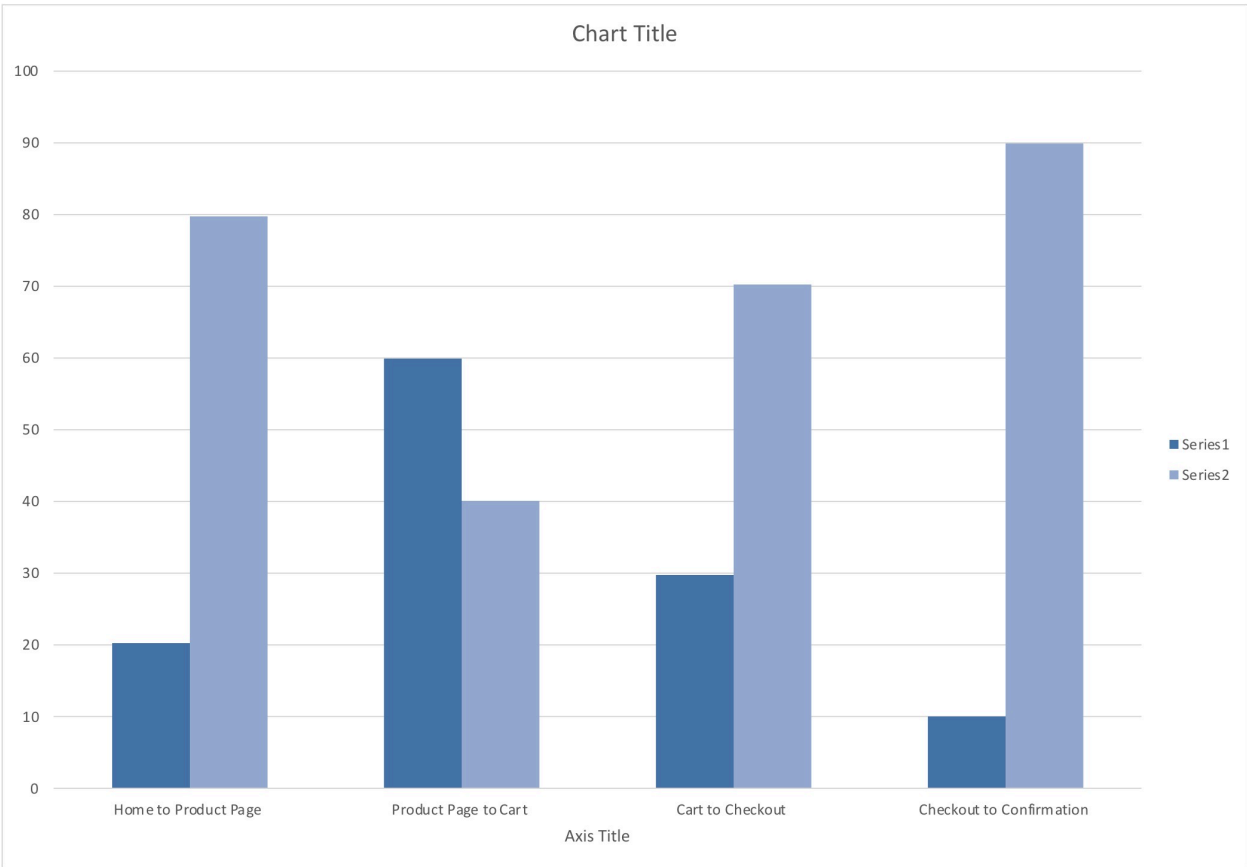
Funnel Overview



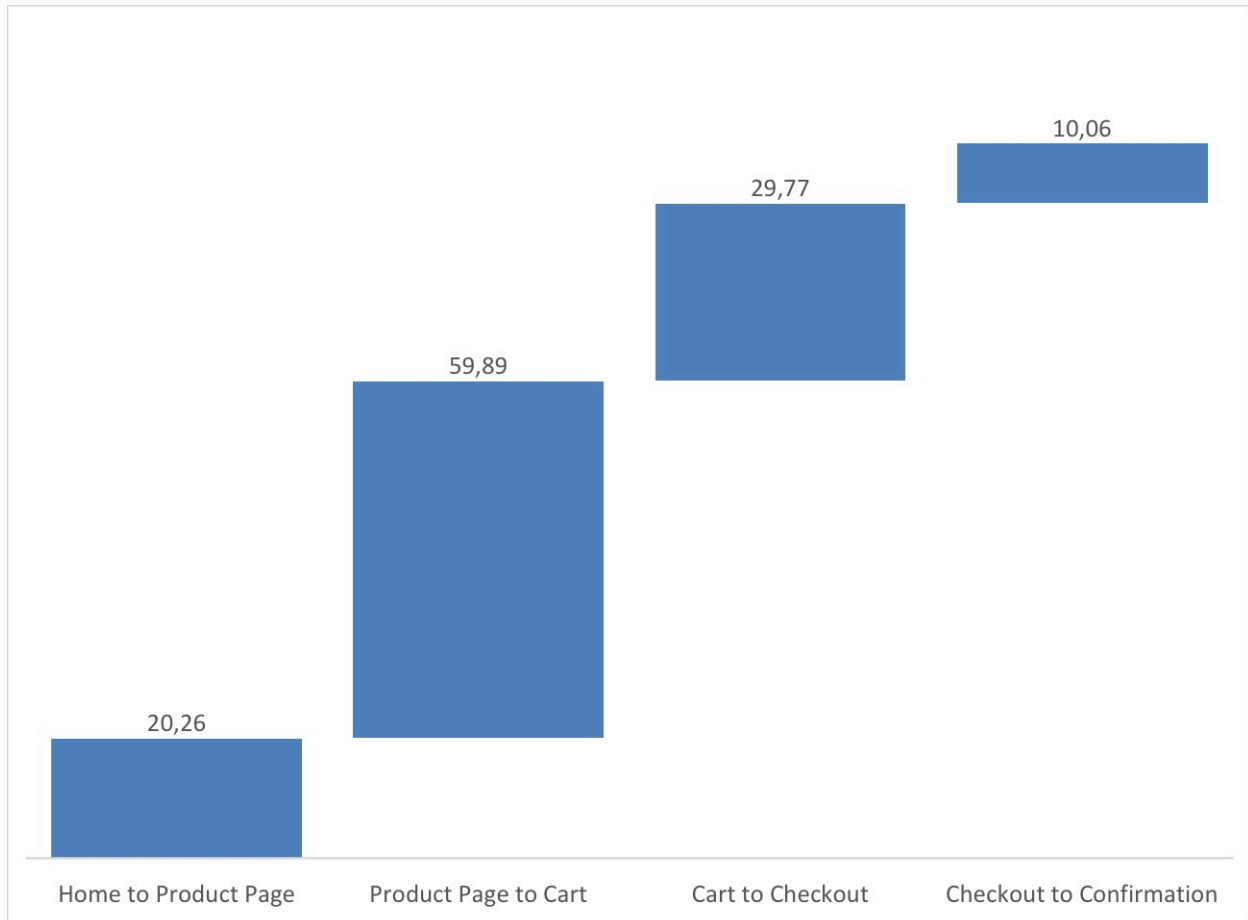
Funnel diagram depicting the number of users retained at every stage of the funnel

Funnel Stages:

Home → Product → Cart → Checkout → Confirmation



Funnel Drop-Off Analysis



 **Bar Chart: Drop-Off Rates by Funnel Stage**

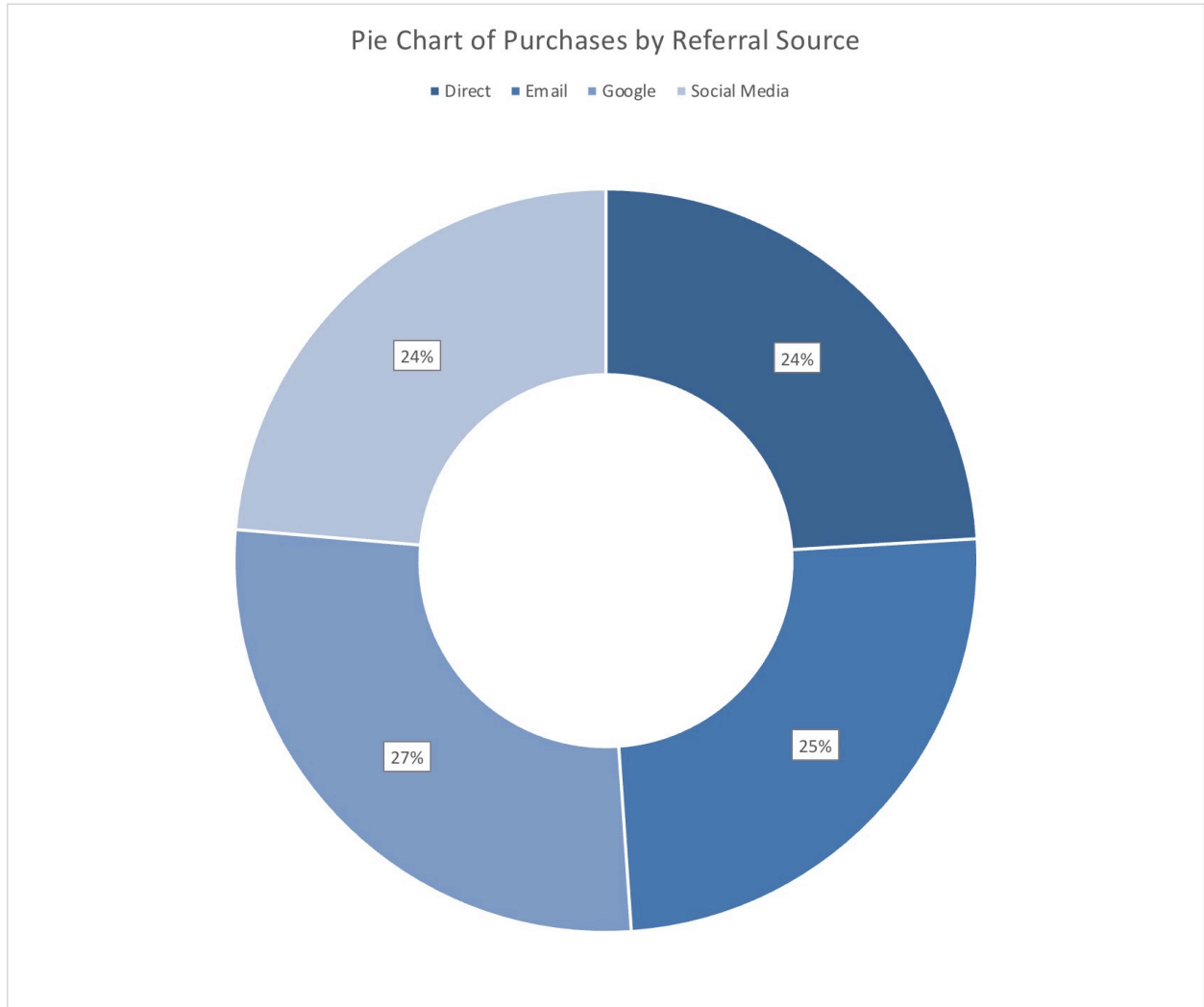
Key Findings

- Approximately **20.2% of users reach the purchase confirmation stage**
- The **Product Page → Add to Cart** stage exhibits the highest drop-off (**~59.9%**)
- Drop-off from Cart → Checkout is moderate
- Checkout completion is relatively strong once reached

Interpretation

Users demonstrate browsing intent but hesitate during product evaluation, suggesting friction related to **pricing clarity, value perception, or UX design**.

Referral Source Analysis



 **Pie Chart: % of Confirmed Purchases by Referral Source**

Traffic Overview

Referral Source	Users	% of Total Purchases
Direct	243	24.06%
Email	251	24.85%
Google	277	27.43%
Social Media	239	23.66%

Insights

- **Google (27.43%)** was the initiator of most completed purchases, followed by Email (24.85%), Direct (24.06%), and Social Media (23.66%).
 - Traffic distribution is relatively balanced, indicating **no single dominant referral source**
 - Balanced traffic suggests that **conversion efficiency**, not volume alone, is the main optimisation lever.
 - This makes **resource prioritisation** (UX vs Development vs Marketing) a suitable optimisation problem.
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Geographic Patterns

Grouped Bar Chart: Drop-Off Rates by Funnel Stage and Continent

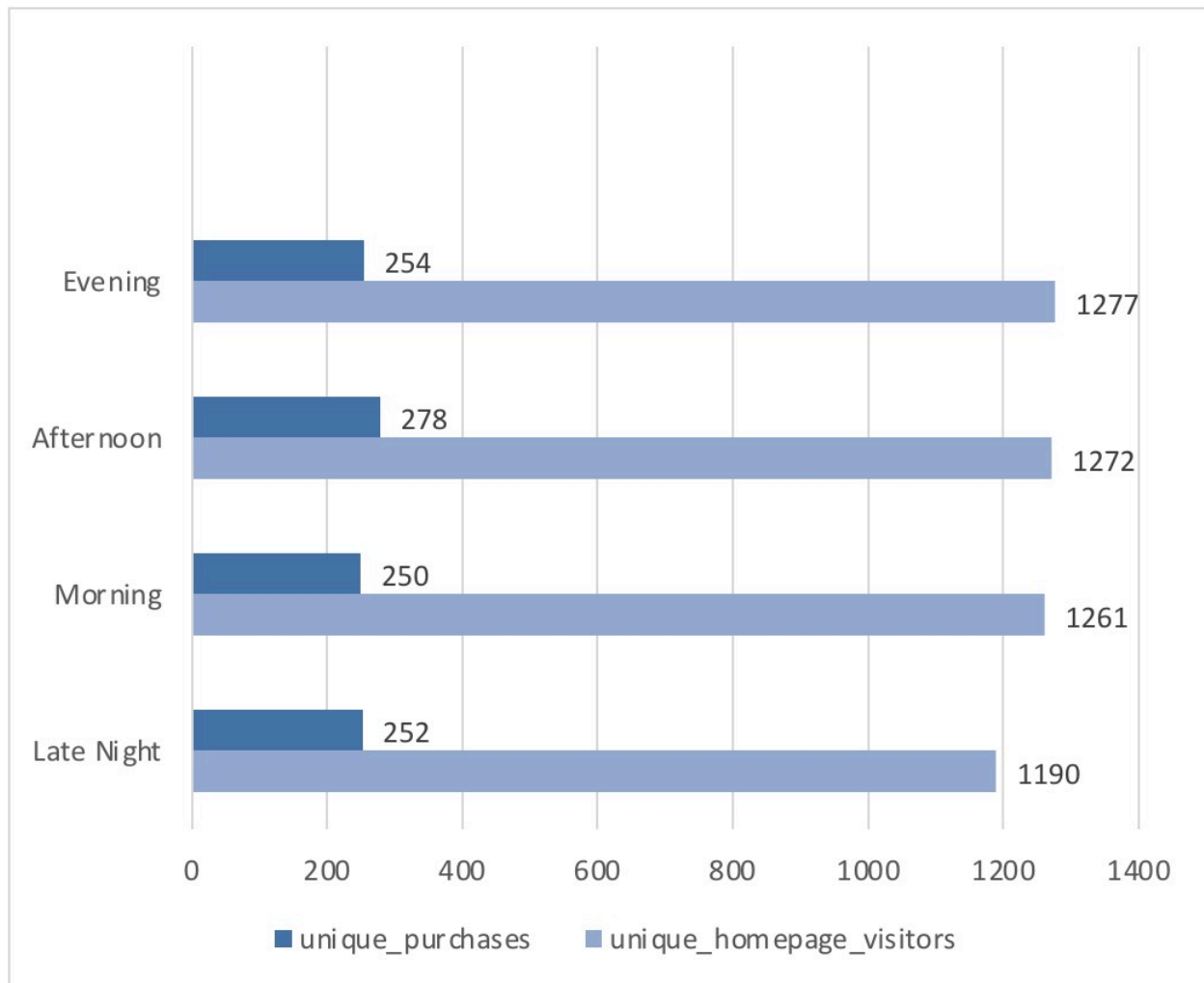
Insights

- Drop-off patterns are broadly consistent across continents
- No single region disproportionately underperforms

Implication

Conversion challenges are **structural**, indicating that global UX improvements should be prioritized over region-specific changes.

Time-of-Day Analysis



 **Bar Chart: User Traffic vs Conversion by Time Segment**

Key Behavioural Insights

- Evening sessions generate the highest traffic volume
- Evening conversion underperforms relative to impressions
- Late-night sessions are fewer but convert more efficiently
- **Afternoon users are most likely to convert**

Opportunity

Promotional activity and engagement nudges such as push notifications should be aligned with **high-intent time windows**.

Key Outcomes & Impact

- Identified the primary funnel bottleneck at the product evaluation stage
 - Confirmed that conversion challenges are structural rather than regional
 - Revealed time-based differences in conversion efficiency
 - Reduced analysis preparation time by **60–70%** through automation
-

Actionable Recommendations

1. Optimize product pages with clearer pricing, value messaging, and trust signals
 2. Improve mobile usability at product and cart stages
 3. Align promotions with high-conversion time windows
 4. Validate improvements through controlled A/B testing
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What I'd Do Next

- Implement automated funnel monitoring dashboards
 - Run A/B tests on product page layout and messaging
 - Track conversion changes following time-based campaign adjustments
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